

# Water Purification and Softening Systems Used in Conjunction with Home Dialysis

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Water Purification and Softening Systems Used in Conjunction with Home Dialysis

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## Description Information

### Benefit Category

Home Dialysis Supplies and Equipment

**Please Note:** This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

### Indications and Limitations of Coverage

#### A. Water Purification Systems

Water used for home dialysis should be chemically free of heavy trace metals and/or organic contaminants which could be hazardous to the patient. It should also be as free of bacteria as possible but need not be biologically sterile. Since the characteristics of natural water supplies in most areas of the country are such that some type of water purification system is needed, such a system used in conjunction with a home dialysis (either peritoneal or hemodialysis) unit is covered under Medicare.

There are two types of water purification systems which will satisfy these requirements:

- Deionization - The removal of organic substances, mineral salts of magnesium and calcium (causing hardness), compounds of fluoride and chloride from tap water using

the process of filtration and ion exchange; or

- Reverse Osmosis (RO) - The process used to remove impurities from tap water utilizing pressure to force water through a porous membrane.

Use of both a deionization unit and RO unit in series, theoretically to provide the advantages of both systems, has been determined medically unnecessary since either system can provide water which is both chemically and bacteriologically pure enough for acceptable use in home dialysis. In addition, spare deionization tanks are not covered since they are essentially a precautionary supply rather than a current requirement for treatment of the patient.

Activated carbon filters used as a component of water purification systems to remove unsafe concentrations of chlorine and chloramines are covered when prescribed by a physician.

## **B. Water Softening System**

Except as indicated below, a water softening system used in conjunction with home dialysis is excluded from coverage under Medicare as not being reasonable and necessary within the meaning of §1862(a)(1) of the Social Security Act. Such a system, in conjunction with a home dialysis unit, does not adequately remove the hazardous heavy metal contaminants (such as arsenic) which may be present in trace amounts.

A water softening system may be covered when used to pretreat water to be purified by a RO unit for home dialysis where:

The manufacturer of the RO unit has set standards for the quality of water entering the RO (e.g., the water to be purified by the RO must be of a certain quality if the unit is to perform as intended);

The patient's water is demonstrated to be of a lesser quality than required; and,

The softener is used only to soften water entering the RO unit, and thus, used only for dialysis. (The softener need not actually be built into the RO unit, but must be an integral part of the dialysis system.)

## **C. Developing Need When a Water Softening System is Replaced with a Water Purification Unit in an Existing Home Dialysis System**

The medical necessity of water purification units must be carefully developed when they replace water softening systems in existing home dialysis systems. A purification system may be ordered under these circumstances for a number of reasons. For example, changes in the medical community's opinions regarding the quality of water necessary for safe dialysis may lead the physician to decide the quality of water previously used should be improved, or the water quality itself may have deteriorated. Patients may have dialyzed using only an existing water softener previous to Medicare end-state renal disease coverage because of inability to pay for a purification system. On the other hand, in some cases, the installation of a purification system is not medically necessary. Thus, when such a case comes to the Medicare Administrative Contractor's (MAC's) attention, the MAC asks the physician to furnish the reason for the changes. Supporting documentation, such as the supplier's

recommendations or water analysis, may be required. All such cases should be reviewed by the MAC's medical consultants.

Cross Reference

The [Medicare Benefit Policy Manual](#), Chapter 15, "Covered Medical and Other Health Services," §110.

## Transmittal Information

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Revision History

05/1989 - Re-categorized dialysis equipment from DME to Home Dialysis Supplies and Equipment for coverage purposes. Effective date NA. (TN 36)

## Additional Information

| Other Versions                                                                  |         |                   |                                                                                       |
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